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Repetitive transcranial magnetic stimulation for nicotine addiction: A regional homogeneity study based on resting-state fMRI

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ABSTRACT

Background: Previous studies have demonstrated the efficacy of transcranial magnetic stimulation (TMS) on treating nicotine addiction. However, the underlying mechanisms remain unclear. This study aims to investigate the effects of repetitive TMS (rTMS) on nicotine addiction using resting-state functional magnetic resonance imaging (rs-fMRI).

Materials and methods: In this study, adult male participants with nicotine addiction were prospectively recruited, and assigned to either the rTMS group ($n = 19$) or Sham group ($n = 12$). Two groups underwent ten sessions of either real or sham rTMS treatment over a two-week period. Clinical assessments related to smoking craving and rs-fMRI scans were conducted before and after treatment. The regional homogeneity (ReHo) was utilized to explore differences in local neural synchronization between the rTMS and Sham groups.

Results: After treatment, the smoking cravings were significantly reduced in two groups. Significant group-by-time interaction effects were observed in the left orbital part of the inferior frontal gyrus, left middle frontal gyrus (MFG), and right angular gyrus (AG). Post-hoc analyses revealed that, compared with pre-treatment, the rTMS group exhibited increased ReHo values after treatment in these areas, while the Sham group exhibited decreased values. Furthermore, within the rTMS group, post-treatment ReHo values in the left MFG were negatively correlated with post-treatment short Tobacco Craving Questionnaire (sTCQ)-Impulse scores. Similarly, the changes in ReHo values of the left MFG from pre- to post-treatment within the rTMS group were negatively correlated with changes in sTCQ-Impulse scores.

Conclusion: Our findings demonstrated that rTMS treatment may improve nicotine-related dependence by modulating local neural synchronization in the prefrontal cortex (PFC) and AG. Furthermore, ReHo values in the left MFG may serve as a promising neuroimaging biomarker for predicting nicotine addiction cessation.

1. Introduction

Tobacco use is one of the most significant global public health threats and a primary cause for mortality among smokers (Le Foll et al., 2022). Globally, approximately 1.3 billion people smoke, with over 8 million fatalities annually due to tobacco-related diseases. Nicotine addiction represents a substantial public health concern worldwide, characterized by craving, loss of control, tolerance, and physical dependence (Fattore and Diana, 2016). Nicotine is a highly addictive chemical in tobacco, which causes considerable harm and makes it difficult for many smokers to quit (Castro et al., 2023; Kolodner et al., 2022). The primary goal for treatment of nicotine addiction is to achieve cessation. Nicotine affects the brain's reward circuitry, reinforcing addictive behaviors through increased dopaminergic activity, which includes the nucleus accumbens (NAc), ventral tegmental area (VTA), amygdala, prefrontal cortex (PFC), locus coeruleus (LC), and hippocampus (Castro et al., 2023; Picciotto and Mineur, 2014). Currently, three therapies, such as varenicline, bupropion, and nicotine replacement therapy, have been approved to be

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effective for smoking cessation, but they only provide short-term benefits, with evidence suggesting that nearly three-quarters of smoker quitters relapse within a year (Hajek et al., 2009; Hajek et al., 2013). Therefore, it is essential to explore diverse strategies for treating tobacco addiction, such as modulating the brain circuits involved in reward, motivation, cognition, and craving control.

Transcranial magnetic stimulation (TMS) is a non-invasive neuromodulation technique based on the principle of electromagnetic induction. By swiftly altering the magnetic field within a coil, TMS generates electric currents capable of penetrating the skull surface without attenuation, thereby directly stimulating the cerebral cortex. High-frequency repetitive TMS (HF-rTMS), operating at frequencies of 5 Hz or greater, enhances neuronal synaptic function, whereas low-frequency rTMS (LF-rTMS), at frequencies below 1 Hz, can suppress neuronal synaptic activity (Klomjai et al., 2015). TMS has been applied in the treatment of various neurological and psychiatric disorders (Cohen et al., 2022), such as depression (Sonmez et al., 2019), obsessive-compulsive disorder (Roth et al., 2021), post-traumatic stress disorder (Koek et al., 2019), schizophrenia (Lorentzen et al., 2022), chronic pain (Lefaucheur, 2006), epilepsy (Tsuboyama et al., 2020) and stroke rehabilitation (Starosta et al., 2022). In recent years, TMS has demonstrated the ability to modulate smoking cravings and nicotine consumption by targeting left dorsolateral prefrontal cortex (DLPFC) [17–21] or insula (Ibrahim et al., 2022; Ibrahim et al., 2019; Ibrahim et al., 2023). Studies have shown that even two sessions of rTMS targeting the left DLPFC in nicotine addicts resulted in reduced smoking and craving (Prikryl et al., 2014). Smokers with damage to the insula are more likely to quit smoking and maintain long-term abstinence (Moran et al., 2012).

Over the past few decades, numerous studies have revealed the abnormal brain alterations associated with nicotine addiction (Lor et al., 2023; Chen et al., 2022; Linli et al., 2023; Fan et al., 2023), alcohol addiction (Crespi et al., 2020; McCalley et al., 2022), and drug addiction (Coray et al., 2023; Lu et al., 2023) using functional magnetic resonance imaging (fMRI) which detects changes in local blood flow and oxygenation related to neuronal activity (Hong et al., 2024; Zhao et al., 2021). Regional homogeneity (ReHo), a reliable metric in resting-state fMRI (rs-fMRI), is commonly used to explore local connectivity and has been widely applied in investigating changes of regional brain activity after TMS (Zhao et al., 2018). Li et al. (2017) found that compared to sham rTMS, active rTMS decreased brain activity in the right insula and thalamus, and reduced the functional connectivity (FC) between the left DLPFC and the orbital middle PFC. Li et al. (2024) also found that rTMS treatment increased the FC between the DLPFC and dorsal anterior cingulate cortex (dACC), and reduced the FC between the dACC and medial orbitofrontal cortex (mOFC). Similarly, Wang et al. (2019) discovered that acupuncture had a significant effect on reducing smoking cravings in nicotine addicts, with the salience network (SN) playing a critical role in this process. However, studies combining fMRI to evaluate the efficacy of rTMS in nicotine addiction treatment remain limited. This study endeavors to investigate the therapeutic efficacy and neural mechanisms underlying rTMS treatment on nicotine addiction utilizing the ReHo approach.

2. Materials and methods

2.1. Participants

This study was conducted at Zhejiang Hospital from March 2022 to December 2024. Adult male participants with nicotine addiction were recruited through posters and online advertisements. The participants were assigned to an rTMS group ($n = 28$) and a Sham group ($n = 15$), with both groups being matched in terms of age and education. Inclusion criteria for the nicotine addiction were as follows: (1) Meeting the criterion of the Diagnostic and Statistical Manual of Mental Disorders V (DSM-V) for nicotine addiction, with daily smoking ≥10 cigarettes and a smoking history ≥2 years; (2) Aged 18–65 years, right-handed male; (3) Motivation to quit smoking. Exclusion criteria for two groups included: (1) History of any brain disease impacting brain function, such as stroke or brain tumor; (2) Organ failure affecting heart or lung function; (3) Inability to undergo MRI, such as claustrophobia; (4) History of epilepsy; (5) Contraindications for MRI; (6) A history of other substance use disorders; (7) Receiving other smoking cessation treatments. Nine subjects in the rTMS group and three subjects in the Sham group were excluded due to non-compliance with the treatment protocol or excessive head movement during fMRI scanning. Finally, 19 participants in the rTMS group, and 12 participants in the Sham group entered into subsequent analysis.

All study procedures were approved by the ethics committee of Zhejiang Hospital (Ethics approval No 2019 Clinical Trial No 34K-X2) and were in accordance with Good Clinical Practice Guidelines and the Declaration of Helsinki. Informed consent was obtained from all participants, with their privacy being fully protected.

2.2. rTMS protocol

The treatment was conducted by three experienced operators using a YRD CCY-II TMS device equipped with a figure-eight coil. rTMS group received 10 sessions of rTMS over two weeks (1 session/day), while the Sham group received 10 sessions of sham rTMS over two weeks (1 session/day) over the same period.

The specific procedure was as follows: (1) Target selection and localization: The target was the left DLPFC. The rTMS coil was first positioned over the motor cortex to determine the optimal site for inducing motor-evoked potentials, after which it was adjusted 2 cm anteriorly to locate the DLPFC. (2) Stimulation parameters: Stimulation was applied at 10 Hz and 100 % intensity, with a continuous stimulation duration of 5 s and an inter-stimulus interval of 20 s, resulting in a total of 1500 pulses delivered over 15 min. During the first treatment session, the resting motor threshold (RMT) was determined by finding the minimum stimulus intensity that elicited activation in the right abductor pollicis brevis muscle in at least 5 out of 10 trials. Treatment was then conducted at an intensity set to 100 % of this RMT, with a positioning cap used for precise localization. The procedures were identical for both groups; however, during the actual treatment, the coil was placed flush against the scalp in the rTMS group to produce a genuine therapeutic effect, whereas in the Sham group the stimulation coil was rotated 90° so that subjects only experienced the sound and vibration, without generating any real therapeutic effect.

2.3. Clinical evaluation and outcome measures

Participants in the rTMS and Sham groups underwent several clinical assessments, including Fagerstrőm Test for Nicotine Dependence (FTND)[21], Heaviness of Smoking Index (HSI) (Borland et al., 2010), Minnesota Nicotine Withdrawal Scale (MNWS)[21], and short Tobacco Craving Questionnaire (sTCQ) (Heishman et al., 2008). These assessments were conducted solely before the initial treatment and at the end of the entire treatment regimen.

2.4. MRI acquisition

MRI scans were performed at a 3.0T superconducting MRI scanner (SIEMENS, 3.0T, Skyra) with a 20-channel phased-array head coil, covering the region from the vertex to the base of the skull. Participants were positioned supine with their heads entering the scanner first, and were instructed to keep their eyes closed and remain awake. Parameters for the 3D-T1 sequence were: 176 slices, slice thickness = 1.0 mm, repetition time (TR) = 1900 ms, echo time (TE) = 2.52 ms, flip angle = 9°, matrix = 256 × 256, imaging time = 4 min and 26 s. Rs-fMRI sequence parameters were: 33 slices, slice thickness = 3.5 mm, TR = 2000 ms, TE = 30 ms, flip angle = 90°, matrix = 64 × 64, imaging time

= 8 min and 8 s. Two MRI scans were performed in the rTMS and Sham groups before the first treatment session and after the final session, respectively.

2.5. MRI data processing

FMRI data were preprocessed using SPM 12 (https://www.fil.ion.ucl.ac.uk/spm) and the RESTplus toolbox (http://www.restfmri.net/forum). The steps included: (1) discarding the first 10 time points; (2) slice time correction; (3) head motion correction (excluding data with head translation > 2.5 mm or rotation > 2.5°); (4) spatial normalization to the Montreal Neurological Institute (MNI) template; (5) linear drift removal; (6) regressing out noise-related variables, including white matter and cerebrospinal fluid signals, and head movement parameters; (7) 0.01–0.08 Hz band-pass filtering; (8) calculating ReHo in the RESTplus and smoothing with a 6 mm full-width half-maximum Gaussian kernel.

2.6. Statistical analysis

Statistical analyses were performed using SPM12 and SPSS 25.0. The Shapiro-Wilk (SW) test was used to assess the normality of quantitative data. Normally distributed data were expressed as mean ± standard deviation (SD), whereas data that did not meet normality criteria were expressed as median (Q1, Q3). Baseline characteristics between the two groups prior to treatment were compared using either two-sample *t*-tests or rank-sum tests. For clinical variables measured before and after treatment in the rTMS and Sham groups, mixed-design analysis of variance (ANOVA) were performed if the data conformed to normality assumptions. If the data violated normality assumptions, within-group comparisons (pre- vs post-treatment) were performed using the Wilcoxon signed-rank tests, and between-group comparisons were conducted using the Mann-Whitney *U* tests. A *P*-value <0.05 was considered statistically significant.

A mixed-design ANCOVA based on the SPM12 was conducted to explore group main effect, time main effect, and group-by-time interaction effect on ReHo values, with age and educational level as covariates. Gaussian Random Field (GRF) correction was applied, with statistical significance set at a voxel-level $P < 0.01$ and a cluster-level $P < 0.05$. Further post-hoc analyses were performed using SPSS 25.0. Specifically, two-sample *t*-tests were used to examine between-group differences, while paired-sample *t*-tests were conducted to assess within-group differences. $P < 0.05$ was considered statistically significant. The ReHo values in the brain regions showing significant differences between two groups were further analyzed correlations with clinical assessments using Spearman correlation analyses, considering $P < 0.05$ as statistically significant.

3. Results

3.1. Baseline characteristics

The rTMS group consisted of 19 right-handed adult males aged 21–54 years (37.47 ± 9.02 years), with 14.84 ± 3.20 years of education. The Sham group consisted of 12 right-handed adult males aged 19–59 years (36.83±13.80 years), with 14.67±3.11 years of education. There were no significant differences in age ($P = 0.888$) and education level ($P = 0.882$) between the two groups.

3.2. Comparison of clinical scale scores

After treatment, the rTMS group exhibited significant decreased in FTND, HSI, MNWS, and sTCQ (Emotion, Expectation, Impulse and Purpose). Similarly, the Sham group exhibited significant decreased in FTND, HSI, MNWS, and sTCQ (expectation, impulse) (all $P < 0.05$, Table 1). However, no statistically significant differences were found between the rTMS and Sham groups when comparing the changes in each clinical scale from pre- to post-treatment (all $P > 0.05$, Table 2). For the MNWS, a mixed-design ANOVA with Group (rTMS vs. Sham) as the between-subjects factor and Time (pre- vs. post-treatment) as the within-subjects factor revealed a significant main effect of Time ($F_{(1,29)} = 27.16$, $P < 0.001$, $\eta^2 = 0.48$), indicating a substantial reduction in withdrawal symptoms across both groups after treatment. Neither the main effect of Group ($F_{(1,29)} = 1.82$, $P = 0.188$, $\eta^2 = 0.06$) nor the group-by-time interaction ($F_{(1,29)} = 0.57$, $P = 0.458$, $\eta^2 = 0.02$) was significant (Table 3).

Table 1
Within-group comparison of clinical scale scores in the two groups.

Comparison	Characteristics	Pre-Treatment	Post-Treatment	*P*
rTMS	FTND	4.00(2.00, 6.00)	2.00(0.00, 4.00)	0.001
	HSI	3.00(2.00, 4.00)	1.00(0.00, 2.00)	0.001
	sTCQ-Emotion	14.00(9.00, 16.00)	6.00(3.00, 9.00)	0.001
	sTCQ-Expectation	17.00(13.00, 19.00)	7.00(3.00, 15.00)	<0.001
	sTCQ-Impulse	9.00(4.00, 15.00)	5.00(3.00, 13.00)	0.002
	sTCQ-Purpose	16.00(13.00, 18.00)	12.00(3.00, 16.00)	<0.001
Sham	FTND	5.50(1.25, 7.75)	1,50(0.25, 5.00)	0.007
	HSI	3.50(1.25, 5.00)	1.00(0.00, 2.75)	0.007
	sTCQ-Emotion	10.50(8.25, 15.75)	8.00(4.25, 14.50)	0.153
	sTCQ-Expectation	11.50(9.00, 19.00)	8.50(4.50, 14.25)	0.036
	sTCQ-Impulse	9.00(7.00, 16.00)	5.50(3.00, 7.50)	0.015
	sTCQ-Purpose	13.00(12.00, 20.50)	7.00(6.00, 15.75)	0.075

Note: FTND: Fagerstrőm Test for Nicotine Dependence, HSI: Heaviness of Smoking Index, MNWS: Minnesota Nicotine Withdrawal Scale, sTCQ: short Tobacco Craving Questionnaire.

3.3. ReHo comparisons between and within groups

Mixed-design ANCOVA revealed no significant main effects for group or time. However, significant group-by-time interactions were observed in the left orbital part of the inferior frontal gyrus (IFG-orb), left middle frontal gyrus (MFG), and right angular gyrus (AG). Post-hoc tests indicated that: (1) the ReHo values in the left IFG-orb and right AG showed no significant difference between the rTMS and Sham groups before treatment, while they were significantly higher in the rTMS group than those in the Sham group after treatment. Specifically, ReHo values increased after treatment in the rTMS group but decreased in the Sham group. (2) the ReHo values in the left MFG were significantly lower in the rTMS group than in the Sham group before treatment, while no

Table 2
Between-group comparisons of clinical scale score changes in the two groups.

Characteristics	Changes in the rTMS group	Changes in the Sham group	*P*
FTND	−2.00(−4.00, −1.00)	−1.50(- 4.75, −1.00)	0.921
HSI	−2.00(−3.00, 0.00)	−1.50(−2.75, −0.25)	0.984
sTCQ-Emotion	−6.00(−11.00, −2.00)	−2.50(−3.75, −0.50)	0.110
sTCQ-Expectation	−5.00(−9.00, −2.00)	−2.00(−8.25, 0.00)	0.152
sTCQ-Impulse	−2.00(−4.00, 0.00)	−5.00(−9.50, −1.25)	0.164
sTCQ-Purpose	−3.00(−7.00, −2.00)	−4.50(−7.50, 2.25)	1.000

Note: FTND: Fagerstrőm Test for Nicotine Dependence, HSI: Heaviness of Smoking Index, MNWS: Minnesota Nicotine Withdrawal Scale, sTCQ: short Tobacco Craving Questionnaire.

Table 3
Mixed-design ANOVA comparison of MNWS between the two groups.

Comparison	Group	Pre-Treatment	Post-Treatment	$F_{(1,29)}$	*P*	*Eta-squared*
	rTMS	11.00 ± 8.90	5.58 ± 4.46			
	Sham	14.83 ± 7.83	7.58 ± 3.92			
group main effect				1.82	0.188	0.06
time main effect				27.16	<0.001	0.48
group-by-time interaction effect				0.566	0.458	0.02

MNWS: Minnesota Nicotine Withdrawal Scale.

significant difference between the rTMS and Sham groups after treatment. Additionally, ReHo values increased after treatment in the rTMS group but decreased in the Sham group (Table 4, Fig. 1A, B and C).

3.4. Correlation between ReHo and clinical scale scores

In the rTMS group, the post-treatment ReHo values in the left MFG were negatively correlated with post-treatment sTCQ-Impulse scores ($R = -0.500$, $P = 0.029$, Fig. 2A). Furthermore, the changes in the ReHo values of the left MFG from pre- to post-treatment were negatively correlated with changes in sTCQ-Impulse scores ($R = -0.510$, $P = 0.026$, Fig. 2B) . No significant correlations were observed in the Sham group.

4. Discussion

In this study, we explored the neural mechanisms underlying rTMS treatment in nicotine addiction by analyzing ReHo. Mixed-design ANCOVA revealed significant group-by-time interaction effects, primarily involving the left IFG-orb, left MFG, and right AG. Furthermore, within the rTMS group, significant negative correlations were observed between post-treatment ReHo values in the left MFG and post-treatment sTCQ-Impulse scores, as well as between the changes in ReHo from pre- to post-treatment and changes in sTCQ-Impulse scores. These findings underscore the potential effectiveness of rTMS in treating nicotine addiction.

Craving is regarded as a pivotal symptom of nicotine addiction and serves as a robust predictor of smoking cessation outcomes (Taggar et al., 2015). The reduction in nicotine dependence scores observed in this study aligns with the previous literature (Abdelrahman et al., 2021; Li et al., 2020; Mikellides et al., 2023; Zangen et al., 2021; Antonelli et al., 2021). However, no statistically significant differences were found between groups regarding changes in these clinical measures. These findings might be attributed to factors such as the limited sample size, placebo effects, and possibly the insufficient duration of rTMS intervention to achieve a robust clinical effect. In the sham group, although the coil did not deliver effective cortical stimulation, we still observed significant reductions in craving scores. This phenomenon can be explained through neurobehavioral mechanisms. According to expectation theory (Lee et al., 2015), the positive anticipations formed during treatment may activate prefrontal-striatal circuits, generating placebo-like effects that reduce subjective craving. Concurrently, based on cue extinction theory (Kearns et al., 2012), the standardized daily treatment protocol constituted systematic nicotine-free exposure to smoking-related cues. Through repetitive conditioned responses, this process likely weakened the original cue-craving neural associations. Notably, during subsequent telephone follow-ups, we observed that some long-term effects persisted in the rTMS group, while improvements in the Sham group dissipated relatively quickly. This suggests a potential long-term benefit of rTMS treatment, though our follow-up sample was limited, and further investigation is warranted. Our study further supported the efficacy of rTMS targeting the left DLPFC in promoting smoking cessation. These behavioral improvements further highlight the promising clinical application potential of rTMS as a non-invasive intervention, particularly in alleviating smoking cravings and withdrawal symptoms. Additionally, in this study, no adverse events were observed in the rTMS group during the treatment period.

The present study observed significant group-by-time interaction effects in the left IFG-orb, left MFG, and right AG. The PFC played a crucial role in the maintenance and progression of nicotine addiction, given its involvement in various neural processes associated with addictive behaviors, including impulse control, decision-making, and reward processing (Tiwari et al., 2020). Yu et al. (2013) observed decreased ReHo in the PFC in nicotine addicts compared to non-smokers.

The left IFG-orb, a crucial node within the PFC, was involved in reward evaluation, emotional regulation, and decision-making predictions (Heather Hsu et al., 2020; Liakakis et al., 2011). Xue et al. (2020)) found reduced dynamic ReHo in the IFG and MFG in nicotine addicts compared to healthy controls, and the temporal variability of dynamic ReHo in the IFG was negatively correlated with the FTND.

Table 4
Mixed-design ANOVA comparison of ReHo values between the two groups.

Comparison	Number	Regions	Voxels	MNI			*F*	*T*	P
				X	Y	Z			
	1	L-IFG-orb	105	−33	21	−15	19.08		
Pre-T vs. Pre-S								−2.07	0.055
Post-T vs. Post-S								2.96	0.006
Post-T vs. Pre-T								3.24	0.005
Post-S vs. Pre-S								−3.79	0.003
	2	L-MFG	330	−39	27	36	27.39		
Pre-T vs. Pre-S								−2.74	0.014
Post-T vs. Post-S								1.84	0.077
Post-T vs. Pre-T								4.18	0.001
Post-S vs. Pre-S								−4.40	0.001
	3	R-AG	68	57	−60	33	13.57		
Pre-T vs. Pre-S								−1.26	0.219
Post-T vs. Post-S								3.32	0.003
Post-T vs. Pre-T								2.19	0.042
Post-S vs. Pre-S								−5.14	<0001

Note: GRF correction was applied with voxel-level $P < 0.01$ and cluster-level $P < 0.05$. T: rTMS group; S: Sham group; MNI: Montreal Neurological Institute; L: left; R: right; IFG-orb: orbital part of inferior frontal gyrus; MFG: middle frontal gyrus; AG: angular gyrus.

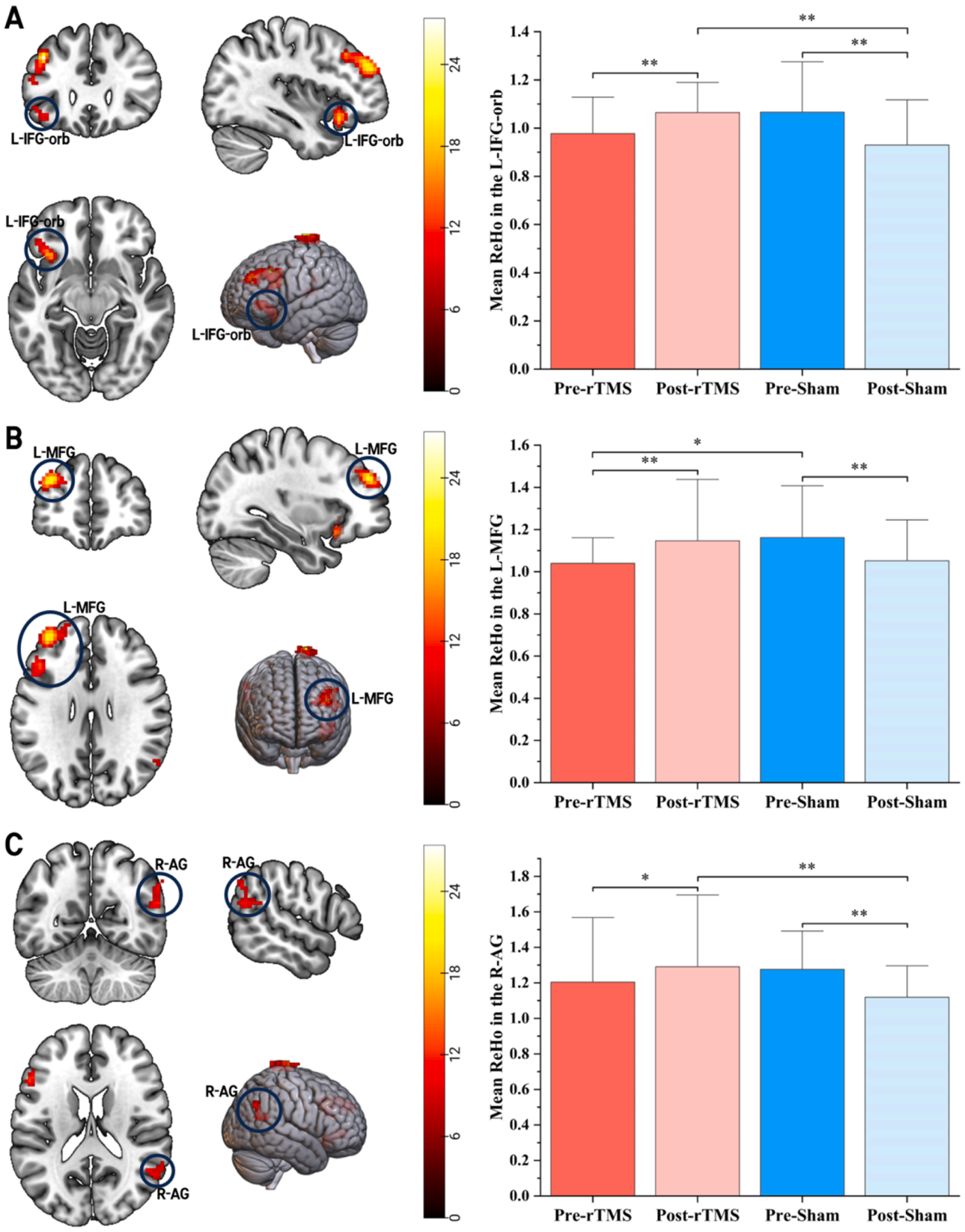


Fig. 1. Brain regions showing significant group-by-time interactions in ReHo between the two groups based on mixed-design ANOVA. The brain regions exhibiting significant group-by-time interactions included the left IFG-orb (Fig. A), left MFG (Fig. B), and right AG (Fig. C). The color bar represents the *F*-values of the comparisons. The bar graph displayed the mean differences in ReHo values across corresponding brain regions under different conditions. *:$P < 0.05$;**:$P < 0.01$;L: left; R: right; IFG-orb: orbital part of inferior frontal gyrus; MFG: middle frontal gyrus; AG: angular gyrus.

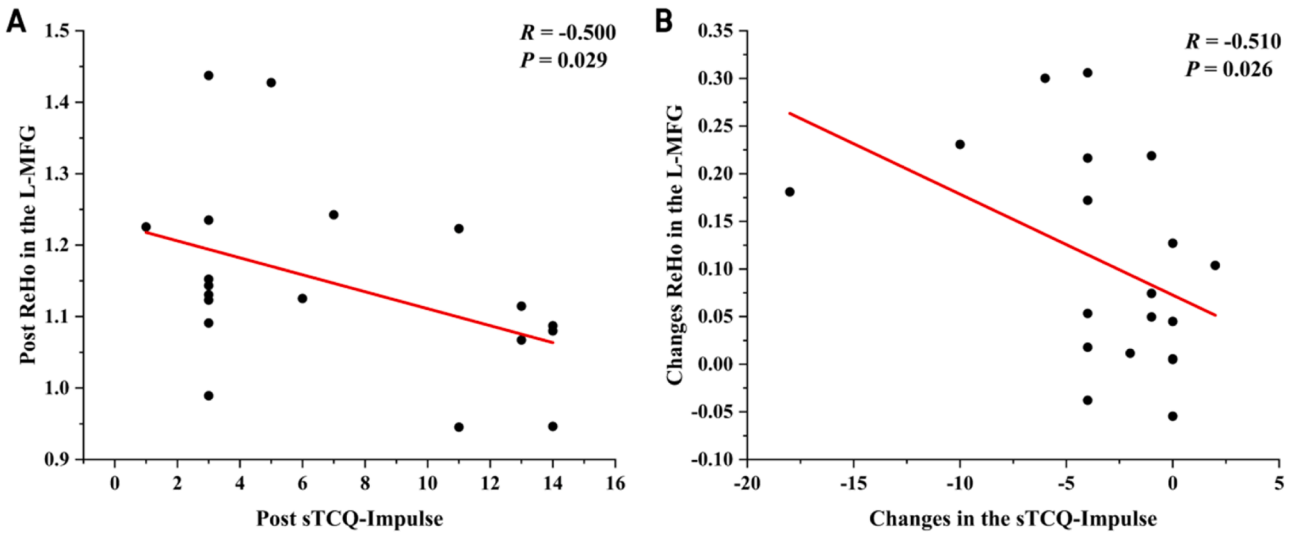


Fig. 2. Correlation between the ReHo and sTCQ-Impulse in the rTMS group. In the rTMS group, the post-treatment ReHo values in the L-MFG was negatively correlated with post-treatment sTCQ-Impulse ($R = -0.500$, $P = 0.029$, Fig. A). The changes in the ReHo values in the L-MFG was negatively correlated with changes in sTCQ-Impulse ($R = -0.510$, $P = 0.026$, Fig. B) . L: left; MFG: middle frontal gyrus; sTCQ: short Tobacco Craving Questionnaire.

Previous studies suggested that abnormal activity in this region could lead patients to provide exaggerated positive feedback to smoking-related stimuli (Paulus, 2007), thereby intensifying nicotine craving and impairing the ability to suppress craving when exposed to smoking cues. In this study, the ReHo values in the left IFG-orb showed no significant difference between the rTMS and Sham groups before treatment, while they were significantly higher in the rTMS group than those in the Sham group after treatment. Specifically, ReHo values increased after treatment in the rTMS group but decreased in the Sham group. The observed improvements in ReHo values following rTMS treatment suggested that rTMS might alleviate abnormal neural functions associated with nicotine addiction through modulating local synchronization within these regions. Specifically, improved function in the left IFG-orb potentially enhanced emotional regulation in response to smoking cues, enabling better inhibition of craving.

The left MFG was primarily involved in executive functions such as working memory, attention, and cognitive control (Niu et al., 2023; Briggs et al., 2021). Zuo et al. (2017) also reported decreased ReHo values in the MFG among nicotine-dependent individuals compared to healthy controls. Dysfunction in this region may reduce self-control capability when encountering smoking-related cues, thus promoting the development of addictive behaviors (Bell and Froeliger, 2021) In this study, the ReHo values in the left MFG were significantly lower in the rTMS group than in the Sham group before treatment, whereas no significant difference was found between the rTMS and Sham groups after treatment. Additionally, ReHo values increased after treatment in the rTMS group but decreased in the Sham group. The observed improvements in ReHo values following rTMS treatment suggested that the increased ReHo in the left MFG might reflect improved executive control and impulse inhibition, allowing individuals to more effectively resist smoking urges.

The right AG was closely related to multimodal sensory integration, attention allocation, and contextual memory processing (Ramanan et al., 2018). Wu et al. (2015) also reported decreased ReHo values in the MFG and AG among nicotine-dependent subjects compared to non-smokers. Abnormal activity in the AG might disrupt the integration of sensory information (such as visual and olfactory cues) associated with smoking, playing an important role in triggering or maintaining nicotine craving (Lin et al., 2020). In this study, the ReHo values in the right AG showed no significant difference between the rTMS and Sham groups before treatment, while they were significantly higher in the rTMS group than those in the Sham group after treatment. Specifically, ReHo values increased after treatment in the rTMS group but decreased in the Sham group. The observed improvements in ReHo values following rTMS treatment suggested that the increased ReHo in the right AG might have strengthened multimodal information integration, supporting healthier cognitive processing of environmental cues associated with addiction and thereby reducing conditioned craving responses.

In summary, the ReHo values of the three brain regions mentioned above significantly increased after treatment in the rTMS group, whereas an opposite trend (decrease) was observed in the Sham group. This divergence suggested distinct underlying mechanisms: whereas active rTMS enhanced local neural synchronization in addiction-related circuits, the Sham group's ReHo reduction may reflect negative neuropsychological effects induced by the ineffective intervention. Specifically, the lack of genuine neuromodulation in sham procedures likely disrupted participants' reward-related expectations, potentially triggering frustration and heightened nicotine craving. These changes were manifested as reduced spontaneous neural activity in key addiction-related regions (e.g., reward or cognitive control circuits). Our findings indicated that in nicotine addiction treatment, psychological suggestion or behavioral expectation alone,without actual neuromodulation, may not only fail to produce therapeutic effects but could exacerbate addiction-related neural dysfunction through negative feedback mechanisms.

Additionally, the present study found negative correlations between the ReHo values in the left MFG and sTCQ-Impulse scores post-treatment in the rTMS group, as well as between the change in left MFG ReHo values and the changes in sTCQ-Impulse scores within the same group. These findings further supported the relationship between improved neural synchronization in the left MFG and reductions in nicotine dependence.

This study has several limitations. First, the sample size was relatively small, which might have led to Type II errors. However, we applied strict inclusion criteria to ensure high data quality and consistency, laying a solid foundation for future large-scale studies. Second, despite the short treatment duration, our study demonstrated the significant short-term efficacy of rTMS on nicotine addiction, verifying its potential for rapid intervention. Previous studies have suggested that extending the rTMS treatment cycle may significantly enhance the suppression of substance cravings (Gay et al., 2022), particularly in addiction treatment. This study provides important evidence for future research. Third, despite the absence of long-term follow-up data, the preliminary short-term follow-up results indicated the efficacy of rTMS

effectiveness in facilitating smoking cessation, laying the groundwork for assessing its long-term effectiveness. Finally, this study was confined to male participants, which may constrain its generalizability to females. Nevertheless, this minimized potential confounding variables related to gender differences, enabling us to focus on the effects of rTMS specifically within the male population. Literature indicates that gender differences may influence efficacy (Slan et al., 2024), and future research will explore its effects in females.

5. Conclusions

In summary, our findings elucidated that the IFG-orb, MFG, and AG represented distinct functional modules closely related to emotional reward evaluation, executive control, and information integration, respectively. By modulating local neural synchronization within these regions, rTMS treatment provided a critical neural basis for alleviating dysfunction associated with nicotine addiction. Furthermore, ReHo values in the left MFG may serve as a promising neuroimaging biomarker for predicting nicotine addiction cessation.

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Informed consent statement

Informed consent was obtained from all subjects involved in the study.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available due to confidentiality.

Ethics approval

All study procedures were approved by the ethics committee of Zhejiang Hospital (Ethics approval No 2019 Clinical Trial No 34K-X2) and were in accordance with Good Clinical Practice Guidelines and the Declaration of Helsinki.

CRediT authorship contribution statement

Zhiqiang Li: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xuan Sha:** Writing – review & editing, Formal analysis, Data curation. **Qiaoli Zhang:** Investigation, Data curation. **Siyin Li:** Investigation, Data curation. **Mei Xie:** Investigation, Data curation. **Tao Wang:** Investigation, Data curation. **Dongyan Chen:** Investigation, Data curation. **Enhui Xin:** Writing – review & editing, Supervision. **Jianjun Zhang:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

Enhui Xin is employed by Department of Research and Development, United Imaging Intelligence Co., Ltd., Shanghai, China. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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